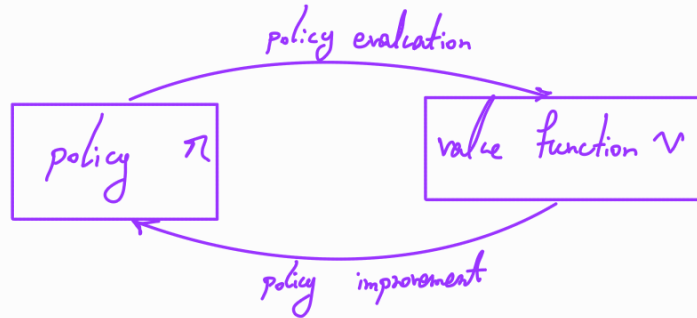


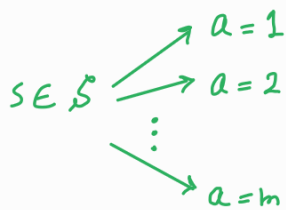
Policy Iteration

PI aims to find an (approximately) optimal policy π^* through iterative policy evaluation and policy improvement.



Policy Improvement: Suppose there exists a deterministic, stationary policy π with value function v^π .

***Question:** Would changing the policy at a state improve the policy?

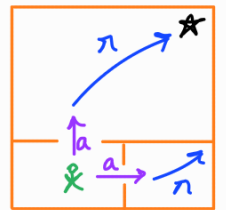


$$\pi(s) = a$$

Would setting $\pi(s) = a' \neq a$ be helpful?

Recall

$$Q^\pi(s, a) = R(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} [v^\pi(s')]$$



If $\exists a \in A$ s.t. $Q^\pi(s, a) > v^\pi(s)$, then changing the policy at state s to a improves the policy.

Note: If for all $s \in S$, $\max_{a \in A} Q^\pi(s, a) = v^\pi(s)$, then

$$v^\pi(s) = v^*(s) \quad \forall s \in S \quad \pi \text{ an optimal policy}$$

Policy Improvement Theorem: Let π and π' be a pair of deterministic, stationary policies such that

$$Q^\pi(s, \pi'(s)) \geq v^\pi(s) \quad \forall s \in S.$$

Then, π' must be as good as or better than π , i.e.,

$$v^{\pi'} \geq v^\pi.$$

Policy Iteration Algorithm

- Initialize $\pi_0: S \rightarrow A$
- For $t = 0, 1, 2, \dots, T-1$.

policy evaluation . Evaluate π_t by computing v^{π_t}

policy improvement . Improve the policy by

$$\pi_{t+1}(s) = \underset{a \in A}{\operatorname{argmax}} \left[R(s, a) + \gamma E_{s' \sim p(\cdot | s, a)} [v^{\pi_t}(s')] \right] \quad \forall s \in S$$

- Return π_T

Lemma: Let v^{π_t} represent the value function of policy π_t at iteration t of PI. It holds that

$$v^{\pi_{t+1}} \geq v^{\pi_t}$$

element-wise $\leftarrow$

Monotonic Improvement

proof :

$$\begin{aligned}
 \forall s \in \mathcal{S} : \quad & v^{\pi_{t+1}}(s) - v^{\pi_t}(s) = R(s, \pi_{t+1}(s)) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi_{t+1}(s))} [v^{\pi_{t+1}}(s')] \\
 & - (R(s, \pi_t(s)) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi_t(s))} [v^{\pi_t}(s')]) \\
 & \geq R(s, \pi_{t+1}(s)) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi_{t+1}(s))} [v^{\pi_{t+1}}(s')] \\
 & - (R(s, \pi_t(s)) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi_t(s))} [v^{\pi_t}(s')]) \\
 & = \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi_{t+1}(s))} [v^{\pi_{t+1}}(s') - v^{\pi_t}(s')]
 \end{aligned}$$

$$\begin{aligned}
 \longrightarrow v^{\pi_{t+1}} - v^{\pi_t} & \geq \gamma P^{\pi_{t+1}} (v^{\pi_{t+1}} - v^{\pi_t}) \\
 & \geq \gamma P^{\pi_{t+1}} (\gamma P^{\pi_{t+1}} (v^{\pi_{t+1}} - v^{\pi_t})) = \gamma^2 (P^{\pi_{t+1}})^2 (v^{\pi_{t+1}} - v^{\pi_t}) \\
 & \vdots \\
 & \geq \gamma^k (P^{\pi_{t+1}})^k (v^{\pi_{t+1}} - v^{\pi_t})
 \end{aligned}$$

$$\text{Let } k \longrightarrow \infty \implies v^{\pi_{t+1}} - v^{\pi_t} \geq 0 \implies v^{\pi_{t+1}} \geq v^{\pi_t}$$

■

Theorem: The value of the final policy v^{π_T} returned by the PI algorithm satisfies $\|v^{\pi_T} - v^*\|_{\infty} \leq \gamma^T \|v^{\pi_0} - v^*\|_{\infty}$.

Proof:

$$\begin{aligned}
 \underbrace{v^*(s) - v^{\pi_t}(s)}_{\geq 0} &= \max_{a \in A} \left[R(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} [v^*(s')] \right] \\
 &\quad - \left(R(s, \pi_t(s)) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi_t(s))} [v^{\pi_t}(s')] \right) \\
 &\leq \max_{a \in A} \left[R(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} [v^*(s')] \right] \\
 &\quad - \left(R(s, \pi_t(s)) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi_t(s))} [v^{\pi_{t-1}}(s')] \right) \\
 &= \max_{a \in A} \left[R(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} [v^*(s')] \right] \\
 &\quad - \max_{a \in A} \left[R(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} [v^{\pi_{t-1}}(s')] \right]
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow \forall s \in S: |v^*(s) - v^{\pi_t}(s)| &\leq \left| \max_{a \in A} \left[R(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} [v^*(s')] \right] \right. \\
 &\quad \left. - \max_{a \in A} \left[R(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} [v^{\pi_{t-1}}(s')] \right] \right| \\
 &\leq \max_{a \in A} \left| \left(R(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} [v^*(s')] \right) \right. \\
 &\quad \left. - \left(R(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} [v^{\pi_{t-1}}(s')] \right) \right| \\
 &= \gamma \max_{a \in A} \left| \mathbb{E}_{s' \sim P(\cdot | s, a)} [v^*(s') - v^{\pi_{t-1}}(s')] \right| \\
 &\leq \gamma \max_{a \in A} \mathbb{E}_{s' \sim P(\cdot | s, a)} [|v^*(s') - v^{\pi_{t-1}}(s')|] \\
 &\leq \gamma \max_{s' \in S} |v^*(s') - v^{\pi_{t-1}}(s')|
 \end{aligned}$$

$$\Rightarrow \max_s |v^*(s) - v^{\pi_t}(s)| \leq \gamma \max_{s \in S} |v^*(s) - v^{\pi_{t-1}}(s)|$$

$$\begin{aligned} \Rightarrow \|v^{\pi_t} - v^*\|_\infty &\leq \gamma \|v^{\pi_{t-1}} - v^*\|_\infty \\ &\leq \gamma^2 \|v^{\pi_{t-2}} - v^*\|_\infty \\ &\vdots \\ &\leq \gamma^t \|v^{\pi_0} - v^*\|_\infty \end{aligned}$$

Value Iteration

vs.

Policy Iteration

- Initialize v_0
- Each iteration:
Apply Bellman optimality operator
- Return v_T & π_T

- Initialize π_0
- Each iteration:
Apply policy evaluation
Apply policy improvement
- Return π_T

search over the space of

value functions, $v \in \mathbb{R}^{|S|}$

deterministic stationary policies, $|A|^{|S|}$ policies

convergence rate of $\|v^{\pi_T} - v^*\|_\infty$

geometric in T

geometric in T

computational complexity

polynomial, $O\left(\frac{|S|^2 |A|}{1-\gamma} \log\left(\frac{1}{(1-\gamma)\epsilon}\right)\right)$

strongly polynomial, $O\left(\frac{|S|^4 |A|^2}{1-\gamma} \log\left(\frac{1}{1-\gamma}\right)\right)$